

### Problem 7.7

At an annual effective interest rate of  $i$ ,  $i > 0$ , the following are all equal:

1. the present value of 10,000 at the end of 6 years;
2. the sum of the present values of 6,000 at the end of year  $t$  and 56,000 at the end of year  $2t$ ; and
3. 5,000 immediately.

Calculate the present value of a payment of 8,000 at the end of year  $t + 3$  using the same annual effective interest rate.

### Solution

Let  $v = \frac{1}{1+i}$  be the annual discount factor.

Since the present value of 10,000 due in 6 years equals 5,000 immediately,

$$10,000v^6 = 5,000.$$

Hence,

$$v^6 = \frac{1}{2}.$$

Also, the second present value equals 5,000, so

$$6,000v^t + 56,000v^{2t} = 5,000.$$

Let

$$x = v^t.$$

Then

$$6,000x + 56,000x^2 = 5,000,$$

which simplifies to

$$56x^2 + 6x - 5 = 0.$$

Using the quadratic formula,

$$x = \frac{-6 \pm \sqrt{6^2 - 4(56)(-5)}}{2 \cdot 56} = \frac{-6 \pm \sqrt{1156}}{112} = \frac{-6 \pm 34}{112}.$$

Since  $x = v^t > 0$ , we take the positive root:

$$x = \frac{28}{112} = \frac{1}{4}.$$

Therefore,

$$v^t = \frac{1}{4}.$$

Now the present value of 8,000 due at time  $t + 3$  is

$$8,000v^{t+3} = 8,000v^t v^3.$$

Since  $v^6 = \frac{1}{2}$ , we have

$$v^3 = \left(\frac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}}.$$

Thus,

$$8,000v^{t+3} = 8,000 \left(\frac{1}{4}\right) \left(\frac{1}{\sqrt{2}}\right) = \frac{2,000}{\sqrt{2}} = 1,000\sqrt{2}.$$

So the required present value is

$$\boxed{1,000\sqrt{2} \approx 1,414.21}.$$